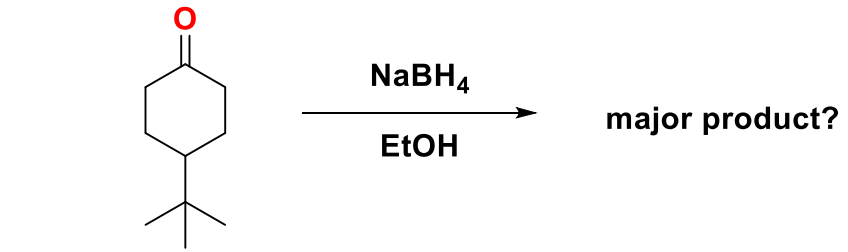
CM3191 Org L3 - Stereoselective Reduction of 4-tert-Butylcyclohexanone



4-t-butylcyclohexanone

Dr. Hoang Truong Giang
chmhoan@nus.edu.sg

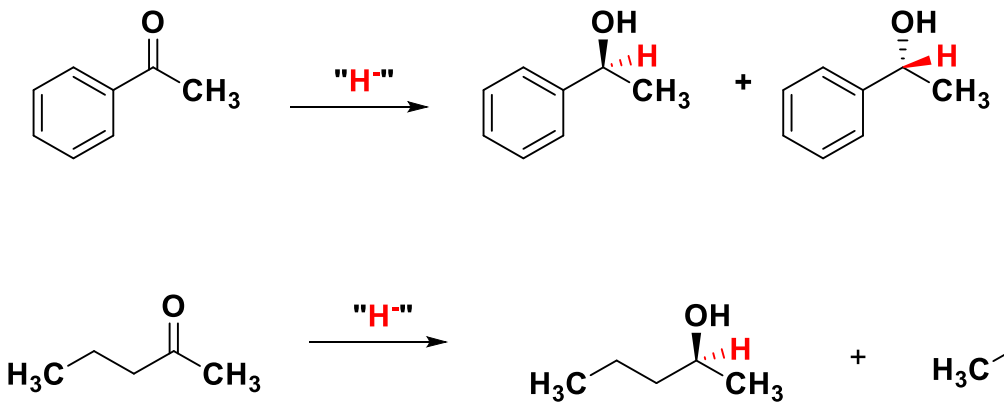
Outline

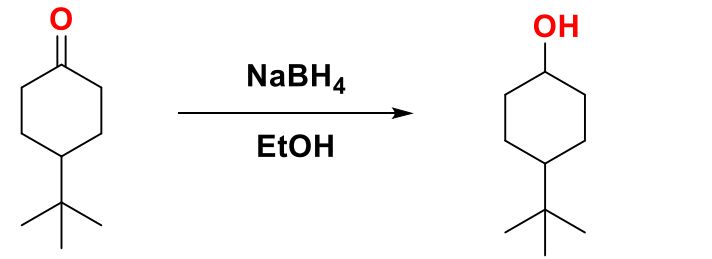
1. Stereoselective Reduction of 4-tert-Butylcyclohexanone
   - Reaction and Procedure
   - Analysis: Relative vs absolute stereochemistry

2. Controlling absolute stereochemistry:
   - Synthesis of Fluoxetine (Prozac): asymmetric reduction
   - Synthesis of Goniothalamin: enzymatic kinetic resolution

Reduction with sodium borohydride

- Sodium borohydride ($NaBH_4$): A mild, hydride-donating reducing agent.
- Scope & Selectivity:
  - Aldehydes → 1°(primary) alcohols
  - Ketones → 2°(secondary) alcohols
- Of the commercially available metal hydrides used for synthetic organic reductions, $NaBH_4$ enjoys the largest industrial use, with an estimated (equivalent) market share greater than 50%.
- A new stereocenter may be generated upon reduction of the ketone.

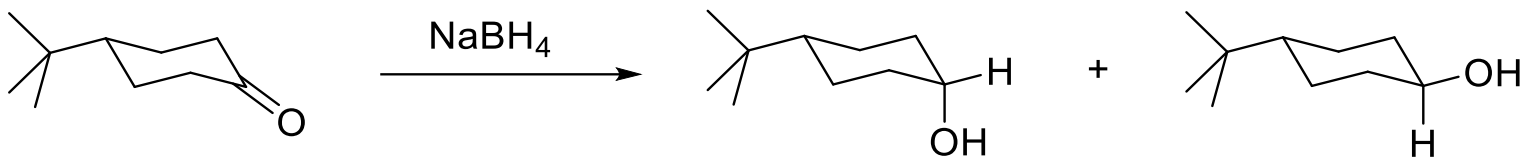




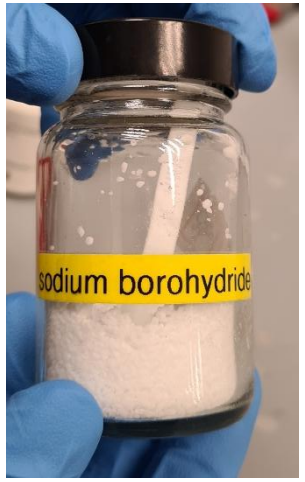
4-t-butylcyclohexanone

4-t-butylcyclohexanol

Reaction scheme



Reagent	MW	mmol	amount
4-tert-butylcyclohexanone	154.25	3.21	0.5 g
sodium borohydride	37.83	10.5	0.4 g
Ethanol	46.07	NA	5 mL
4-tert-butylcyclohexanol	156.27	___	___



white crystalline solid

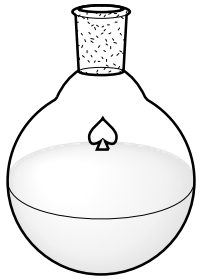
Reactions with $NaBH_4$ are commonly carried out in **protic solvents**:

Methanol: most reactive/efficient

Ethanol: safer alternative

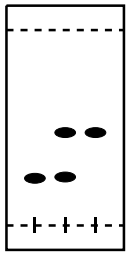
$NaBH_4$ *decomposes in protic solvents:* $NaBH_4 + x\,ROH \rightarrow NaBH_{4-x}(OR)_x + x\,H_2$

Procedure



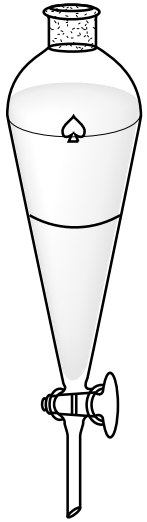
add $NaBH_4$
in small portion

monitor the reaction
progress by TLC



- 4-tert-butylcyclohexanone
- ethanol
- stir bar

- add water
- add HCl
- heat to boiling



Extraction (ethyl acetate)

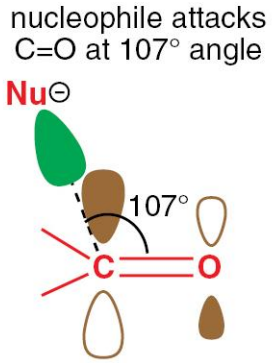
- Dry with Na_2SO_4
- filter
- rotavap

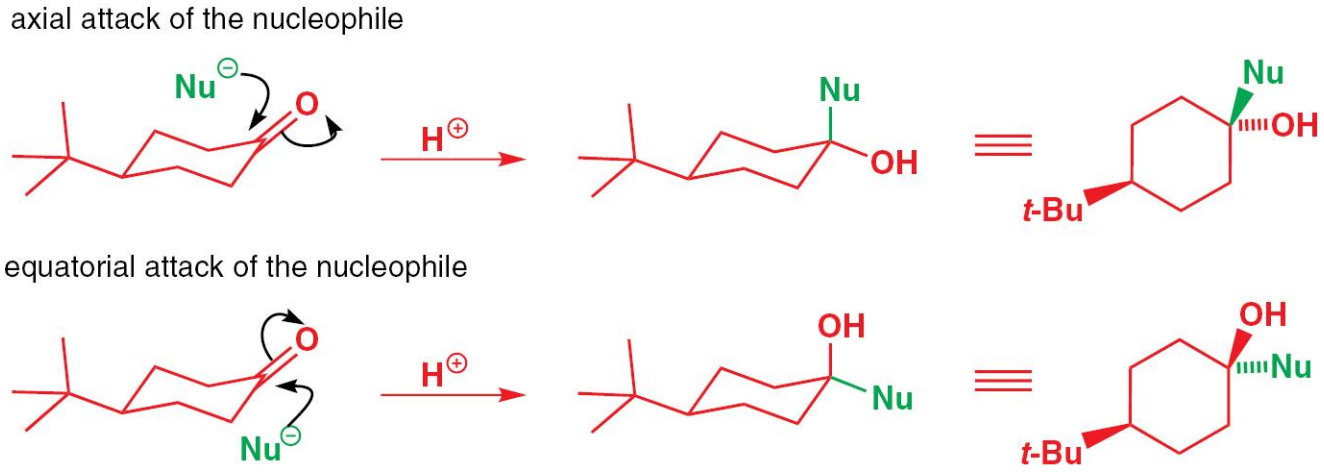
crude product

- yield
- IR
- ^{1}H NMR

Stereoselectivity: cyclohexanone

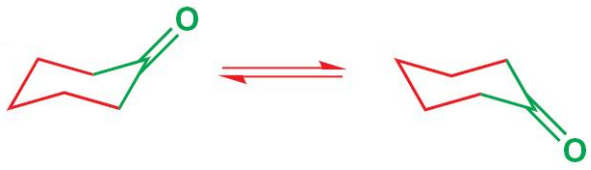
In nucleophilic additions to a carbonyl group, the nucleophile approaches along the **Bürgi–Dunitz trajectory** (107°).



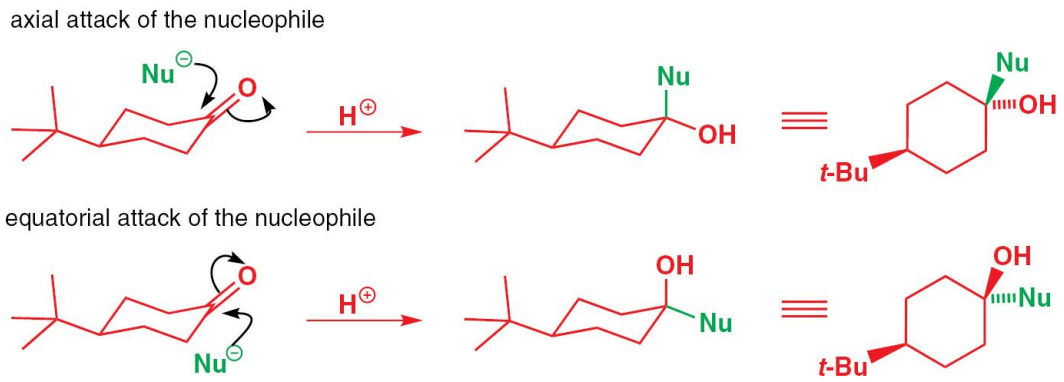


Nucleophilic attack on the cyclohexanone carbonyl can occur via an **axial** or **equatorial** trajectory.

Conformational locking: The bulky *t*-Bu group anchors the cyclohexane ring into a rigid chair conformation.



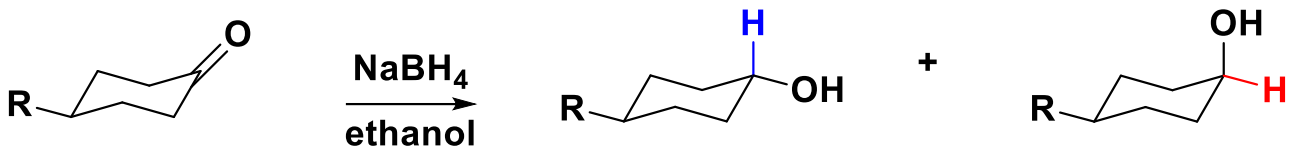
Axial vs equatorial attack (relative stereochemistry control)



Nucleophile	% of product resulting from Axial attack	Equatorial attack
HC≡CLi	88	12
MeLi	35	65
PhLi	42	58
MeMgBr	41	59
EtMgBr	29	71
i-PrMgBr	18	82
t-BuMgBr	0	100

Reference: Clayden 2nd ed, page 828

Result analysis



R = t-Bu

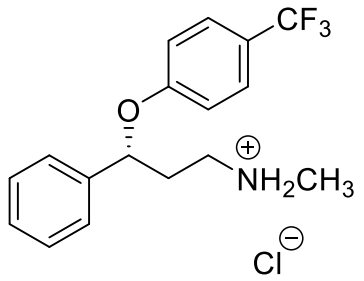
1H NMR	Axial H 3.5 ppm	Equatorial H 4.0 ppm

- ^{1}HNMR: Peak integrations reflect the relative molar ratio of the diastereomers. The remaining signals from the cyclohexane ring protons will likely overlap with one another.

- In your IR spectrum, identify the presence or absence of key peaks to confirm reaction completion. Correlate these findings with your TLC result.

Controlling absolute stereochemistry
Synthesis of Fluoxetine (Prozac)

Catalytic Asymmetric Reductions.



Reading: Corey, E. J.; Reichard, G. A. Enantioselective and Practical Synthesis of *R*- and *S*-Fluoxetines. *Tetrahedron Lett.* **1989**, *30* (39), 5207–5210.

Fluoxetine (Prozac): Impact & Evolution

Therapeutic Uses

Major Depression: First-line adult and pediatric Selective Serotonin Reuptake Inhibitor.

OCD & Panic: High-affinity serotonin transporter (SERT) blockade ($K_i = 17$ nM).

Bulimia Nervosa: First FDA-approved drug therapy.

Discovery & History

1971–1974: Synthesized at Lilly (candidate LY110140).

1987 Approval: Replaced cardiotoxic TCAs and MAOIs.

2001 Patent Cliff: Rapid transition to generic formulations.

Commercial Scale

$2.8B Peak Sales: Record annual revenue in 1998.

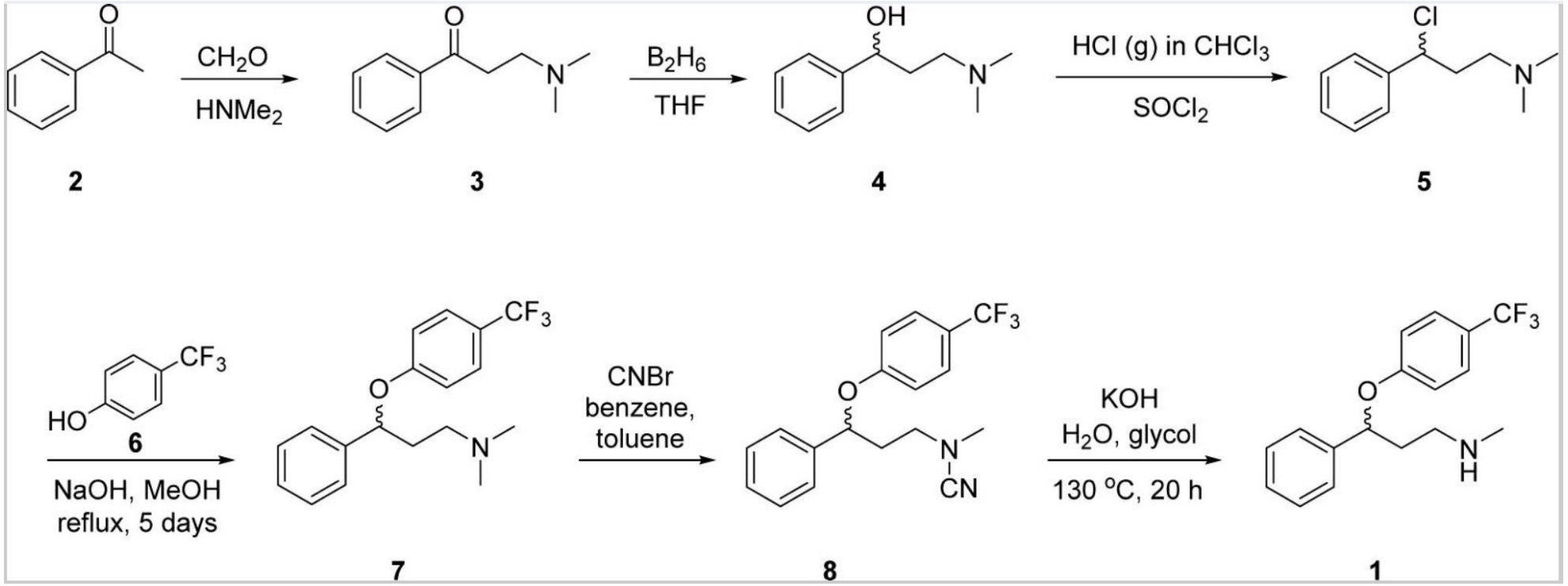
$22B+ Lifetime: First true neuropsychiatric blockbuster.

2023 US Clinical Impact: 18th most prescribed overall drug; 4th most prescribed antidepressant (>27 million prescriptions).

In November 2025, the company reached a $1 trillion market capitalization, the first health-care company in the world to do so

https://en.wikipedia.org/wiki/Fluoxetine
https://en.wikipedia.org/wiki/Eli_Lilly_and_Company

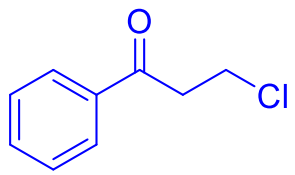
Original Lilly Synthesis of Racemic Fluoxetine



Prozac is sold as a racemate. Both enantiomers inhibit SERT similarly, but (S)-norfluoxetine is ~20–50× more selective for SERT over NET (Norepinephrine Transporter) compared to (R)-norfluoxetine.

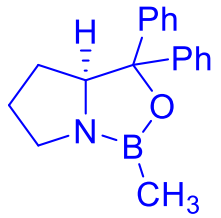
ACS Chem Neurosci. 2013 Nov 9;5(1):14–23. doi: 10.1021/cn400186j

Corey and Reichard's synthesis (1989)

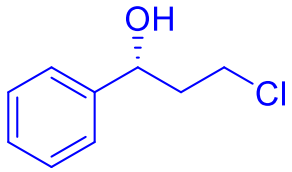


2

0.6 eq BH_3



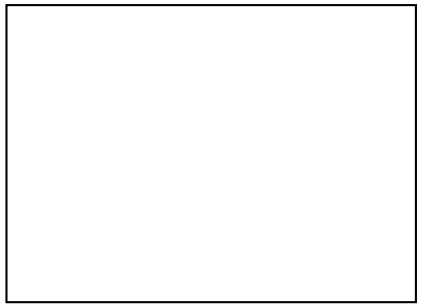
(cat)



4

99% yield
94% ee

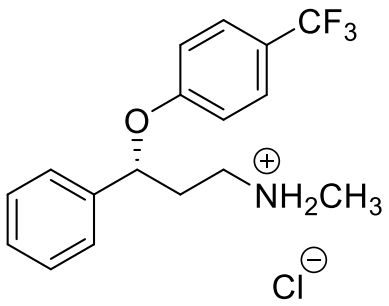
NaI



5

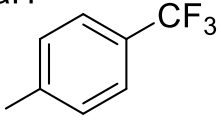
99% yield

CH_3NH_2
(50 eq)

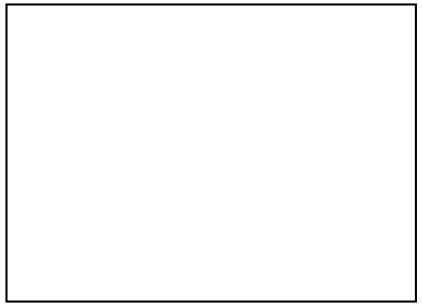


1
Fluoxetine
96 % yield, 99.8% enantiomerically pure

1. NaH
2.



3. HCl

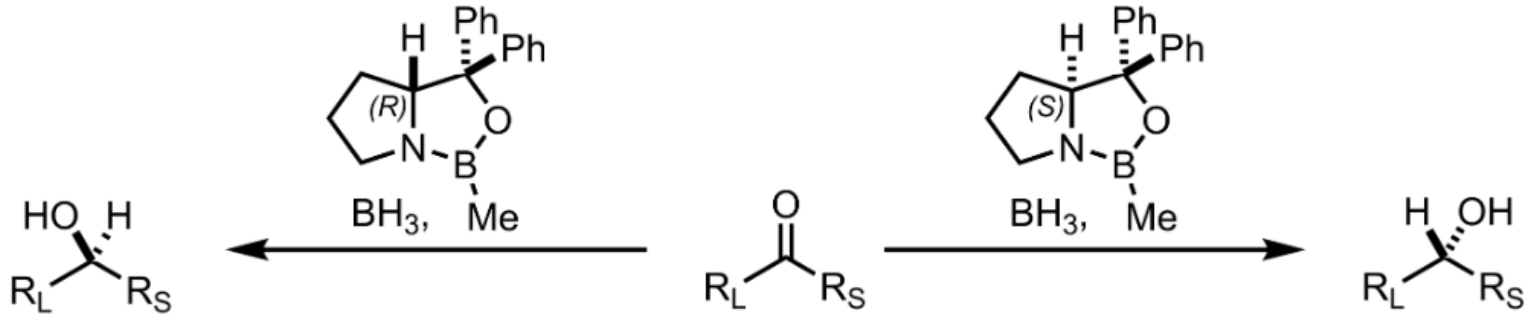


6

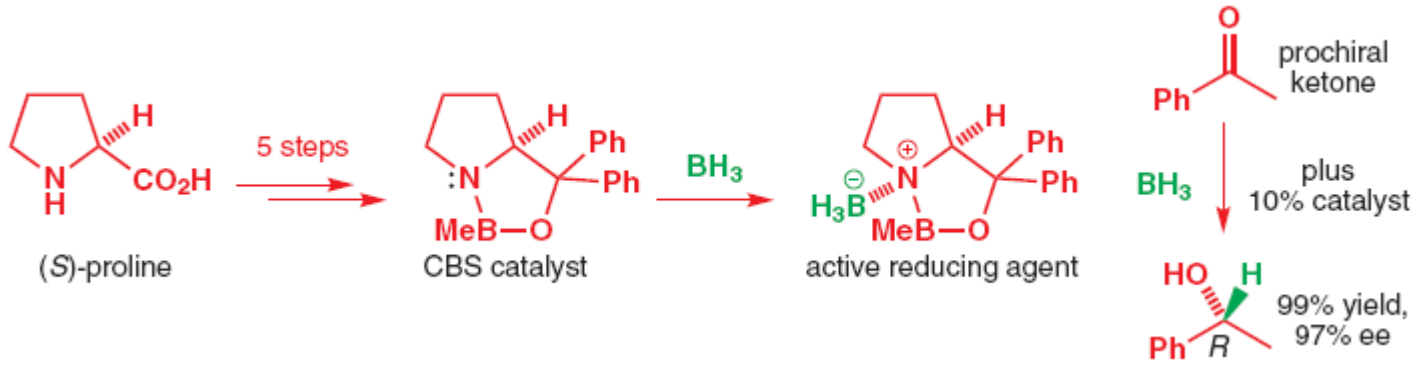
99% yield

Corey–Bakshi–Shibata (CBS) Reduction

Asymmetric Reduction of Prochiral Ketones: Converts prochiral ketones into enantioenriched 2° alcohols with high enantioselectivity (*ee*).



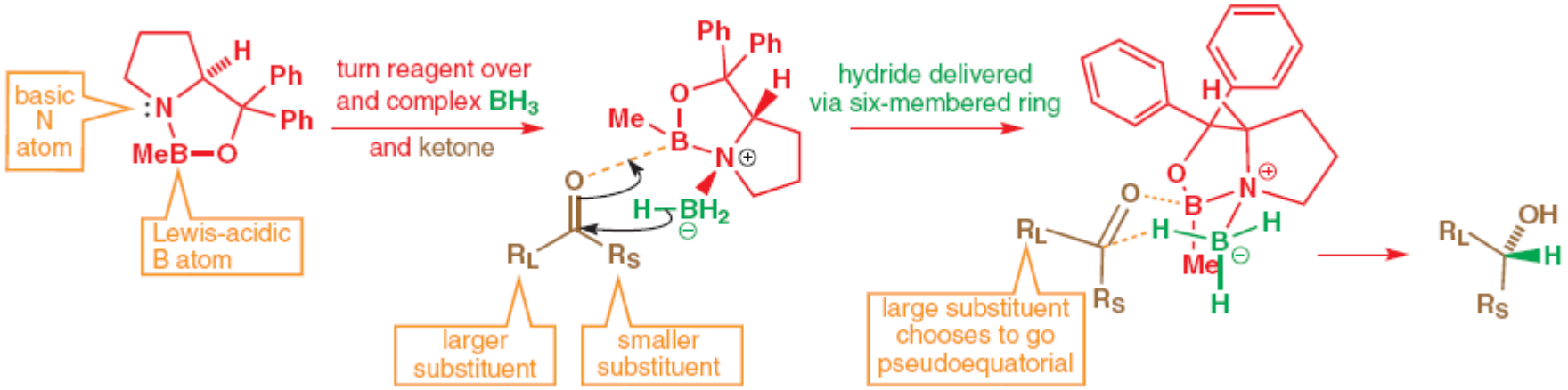
Reagent System: Reducing Agent: Borane complex ($BH_3 \cdot THF$ or $BH_3 \cdot SMe_2$) as the stoichiometric hydride source. Catalyst: Chiral oxazaborolidine (CBS catalyst, derived from L- or D-proline), typically used in sub-stoichiometric amounts (5–10 mol%).



https://synarchive.com/named-reactions/itsuno-corey-reduction

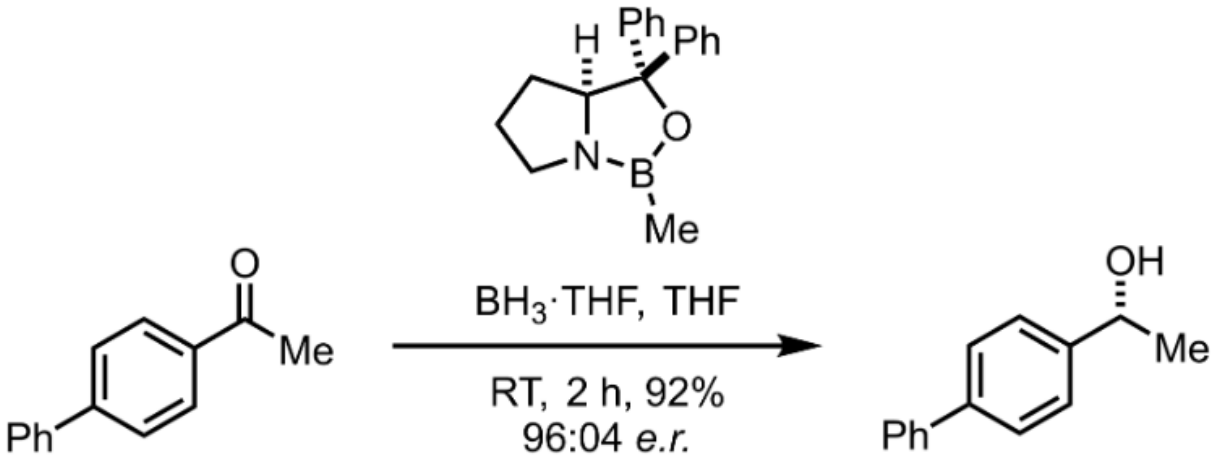
Corey–Bakshi–Shibata (CBS) Reduction

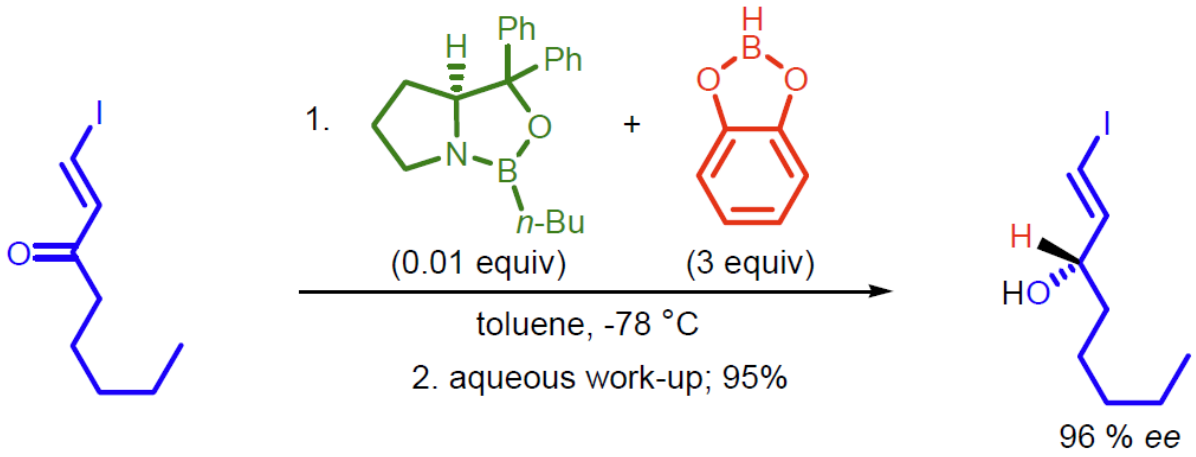
Mechanism



https://synarchive.com/named-reactions/itsuno-corey-reduction

Examples of CBS reduction





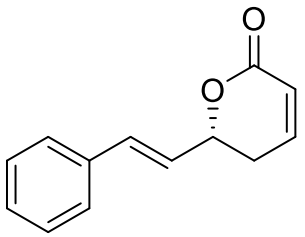
- **Robust Catalyst Handling**: The oxazaborolidine catalyst exhibits good air and moisture stability (often commercially available and shelf-stable as a solid or solution).
- **High Efficiency & Selectivity**: Rapid reaction rates with typically short reaction times. Delivers high chemical yields and excellent enantiomeric excess (often $> 90–95\%$ ee).
- **Predictable Stereocontrol**: Switching between the (R)- and (S)-catalyst enantiomers selectively provides access to either product enantiomer with equal fidelity.

Controlling absolute stereochemistry
Synthesis of Goniothalamin

Kinetic Resolution of Racemic Alcohols

Goniothalamin

- Goniothalamin was first isolated in 1967 from the bark of Cryptocarya caloneura (native to Asia). Originally assigned the (S)-configuration, its absolute stereochemistry was later revised to (R).
- Only the (R)-enantiomer appears in nature.
- (R)-Goniothalamin exhibits potent in vitro cytotoxicity against multiple cancer cell lines, such as MDA-MB-231 (breast), HeLa (cervical), HGC-27 (gastric), HL-60 (leukemia), and Caov-3 (ovarian), while showing minimal toxicity toward non-malignant cells.

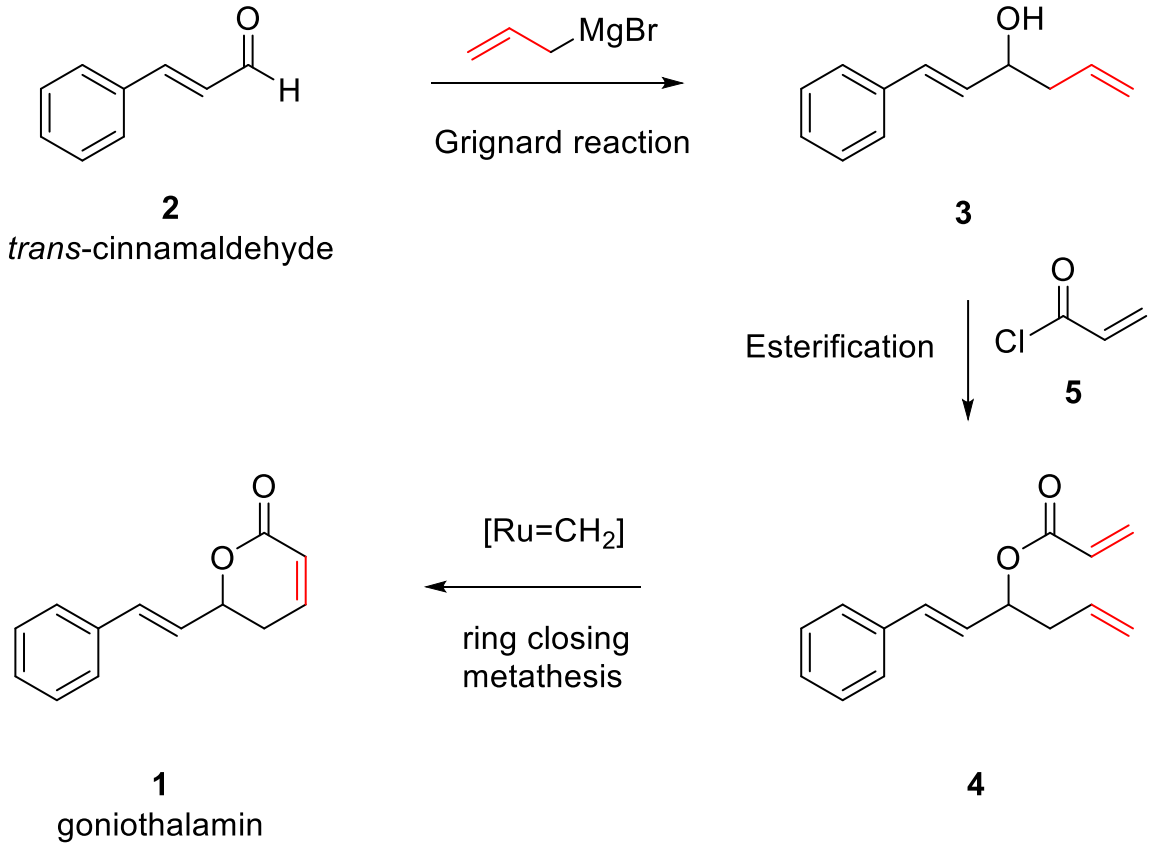


1
(R)-goniothalamin

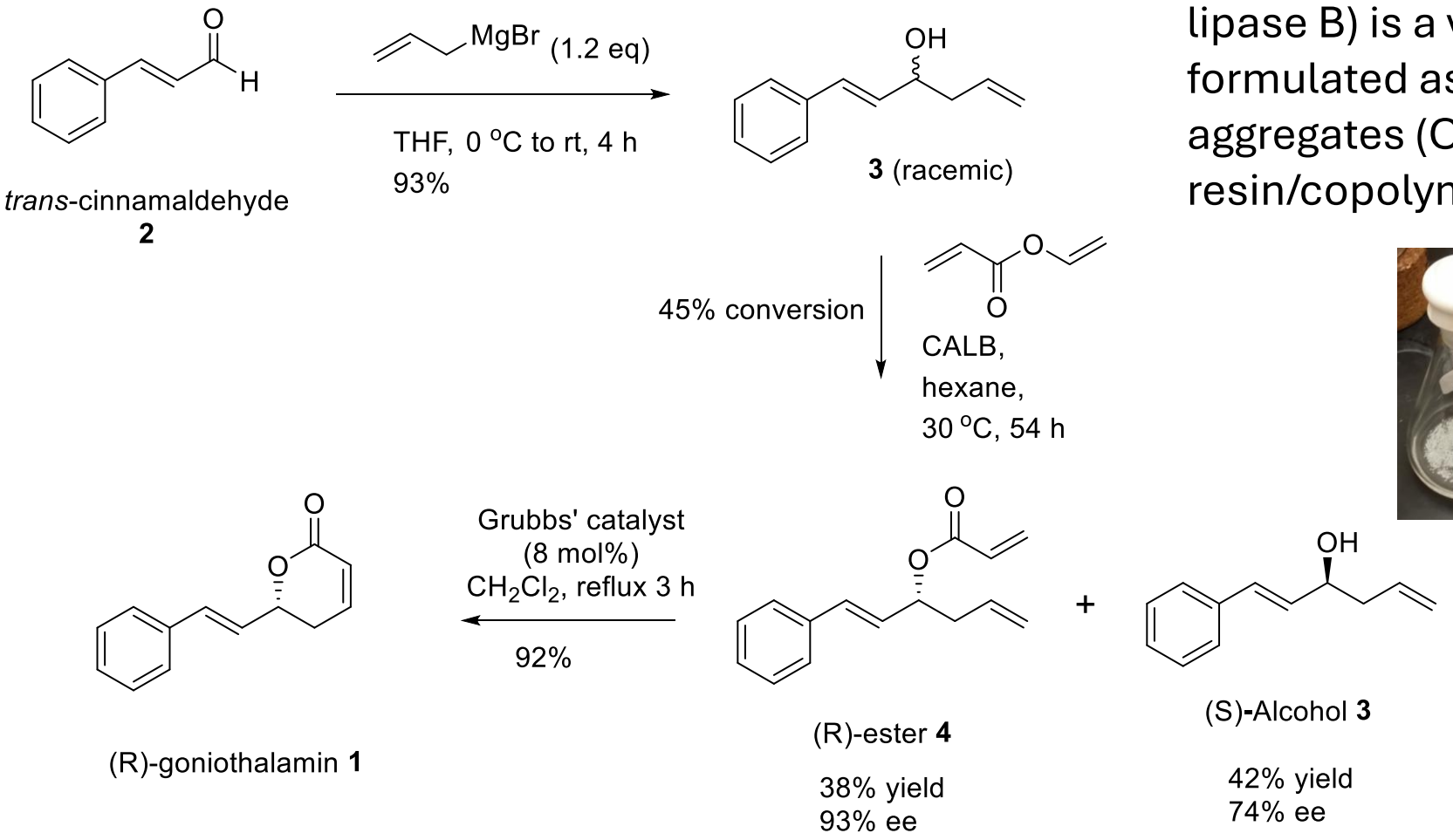
Synthesis of Goniothalamin

3 steps:

1. Grignard reaction
2. Esterification
3. Ring closing metathesis



Synthesis of (R)-Goniothalamin



Immobilized CALB (*Candida antarctica* lipase B) is a versatile hydrolase formulated as cross-linked enzyme aggregates (CLEAs) on a macroporous resin/copolymer support.





incubator shaker

Reference: Anthonsen and Hansen, Tetrahedron **2004**, *60*, 521-524.

Chiral resolution

- Chiral resolution is a process for the separation of racemic compounds into their enantiomers. It is an important tool in the production of optically active drugs.

- Kinetic resolution: Two enantiomers can react at different rates with a chiral entity. The chiral entity may be a biocatalyst (enzyme or a microorganism) or a chemocatalyst (chiral acid or base or even a chiral metal complex).

- Lipase is a commonly used enzyme to perform chiral resolution for alcohols.

  Reaction: Transesterification using an activated ester as an acyl donor.
  Acyl donors: Enol esters such as vinyl acetate or isopropenyl acetate are typically preferred.

